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A Design of Experiment to Evaluate the Printability for Bioprinting by Using Deep Learning Image Similarity

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ABSTRACT

Bioprinting is a growing area in the field of tissue engineering that offers a potential solution to the global shortage of organ transplants. Ensuring high printability is crucial for bioprinting. To better understand printability, a design of experiment model that examines printing speed and pressure in extrusion-based printing was developed. Two biomaterials, hyaluronic acid and sodium alginate, were selected as surrogate biomaterials to understand how rheological properties play a role in printability. Various rheological aspects such as shear-thinning behavior, viscosity, and recovery were investigated. To further evaluate printability, a new method was used that includes deep learning image similarity. The information obtained with the surrogate bioinks was then applied to another biomaterial, methacrylated hyaluronic acid, in combination with corneal keratocytes to demonstrate the successful implementation of the outcome of this design of experiment. As a result of this study, a better understanding of the rheological properties for bioprinting was achieved, leading to a next step towards improving extrusion-based bioprinting, which can be used for a wide range of applications.

1 | Introduction

Over the past decade, bioprinting has emerged as a rapidly advancing technology with rising interest due to its potential to be disruptive and revolutionary. Similar to traditional additive manufacturing such as 3D printing, it is now possible to print biological material in the *x*, *y*, and *z* directions. Bioprinting can be described as the assembly of living and non-living material using a computer-aided transfer process in a predetermined 2D or 3D shape to create bioengineered scaffolds, which can be used in pharmacokinetic, cell biology, tissue engineering, and regenerative medicine [1]. One main goal of bioprinting is to fabricate scaffolds that mimic the functions of human tissues and even organs [2]. It has the potential to reduce the worldwide shortage of organ donors, as it could be an alternative to create transplantable organs. Especially more simple and avascular organs such as the cornea or skin grafts could be made by bioprinting in the near future [3, 4]. Furthermore, great success has been achieved in the printing of functional blood vessels, setting the next step for more complex organs, such as the heart [5, 6].

Unlike 3D printing, where materials such as plastics and metals are used, bioprinting involves the use of soft materials, often in combination with cells. Materials used in bioprinting can be divided into synthetic, semisynthetic, and natural biomaterials. Synthetic materials include materials, for example, such as polyethylene glycol, pluronic, polylactic acid, and polycaprolactone [7, 8]. Common natural biomaterials can be collagen, gelatin, agarose, alginate (ALG), silk, hyaluronic acid (HA), fibrin, chitosan, and decellularized extracellular matrix [9]. Alginate is one of the most common biomaterials, one reason being inexpensive and biocompatible [10]. It is a polysaccharide consisting of ß-D-mannuronic acid and alpha-L-guluronic acid. The percentage of each saccharide determines the gel formation

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and flexibility. A higher percentage of ß-D-mannuronic leads to greater flexibility, while a higher percentage of alpha-L-guluronic acid leads to greater gel formation [11]. Alginate gels can be crosslinked by divalent cations such as Ca^{2+} to form stable scaffolds. Biomaterials are often combined to improve many properties. In this study, alginate is combined with another common biomaterial, hyaluronic acid, being one of the most prevalent components of the extracellular matrix [12]. The combination of ALG and HA serves in this study as surrogate materials to understand the rheology of gels in terms of their printability. However, the combination of ALG and HA has already been used in bioprinting [13].

Numerous different bioprinting technologies exist, with the most common being extrusion printing, drop-on-demand or inkjet, and the increasingly popular SLA (stereolithography) or DLP (digital light printing) [14]. However, extrusion printing still plays a major role in the bioprinting field. Extrusion printing can be divided into pneumatic and mechanical extrusion, while pneumatic extrusion is the most dominant technology [15]. Pneumatic printing is considered to be more cell-friendly than extrusion printing by mechanical force [16]. In this study, a commercial bioprinter (*Bioscaffolder, Gesim*) with pneumatic extrusion was used.

In extrusion printing, printability is the ability to form a scaffold using bioink that matches computer-aided design [17, 18]. To assess printability, there are many different methods, the most widely used focus on pores and strand diameter [19]. The majority of the assessments involve the measurement of the printed structure. This process can be tedious and time-consuming. Furthermore, these assessments often do not consider the whole printed scaffold. Here, a novel method to determine printability was established by using an image comparison of the printed structure with the computer-modeled design (CAD) and a polycaprolactone (PCL) scaffold.

Artificial intelligence, machine learning, and deep learning technologies have been on the rise in the past decade. Especially deep learning and the use of convolutional neural networks (CNN) to classify images have changed the problem of computer vision [20]. Although CNNs have been used in face recognition, human pose estimation, and object detection, they can also be used for image similarity [21]. Cosine similarity was chosen because it is one of the most widely used methods due to outperforming other methods for measuring image similarity [22]. Thus, in this study, three different pre-trained neural network architectures (ResNet50, VGG19, and EfficientNet) were used for image similarity.

To better understand how each hydrogel behaves in extrusion bioprinting, evaluating rheological properties is crucial. Rheological properties such as storage modulus, shear thinning performance, stability behavior, and recovery behavior provide valuable information on the requirements for printability [23]. To this end, various rheological evaluation methods were conducted, including rotational and oscillatory methods.

A design of experiment (DoE) was established using concentrations of HA and ALG as well as printing speed and pressure as varying factors on the printability. A DoE was chosen here as it is an efficient method for optimizing factors. However, the interaction effects and predictive modeling were considered very valuable. Insights were gained on the correlation between printing speed and pressure.

The goal of the study was the implementation of image similarity as an alternative to current measurements of printability. Furthermore, an ALG/HA hydrogel was used as a surrogate hydrogel to understand the rheological properties and improve printability. The DoE statistical analysis revealed a correlation between printing speed and pressure. To prove this DoE model, a bioprint with a methacrylated hyaluronic acid (HAMA) bioink with immortalized corneal keratocytes was conducted. To further demonstrate cell viability, live/dead staining was performed over the course of 1 week.

2 | Materials and Methods

2.1 | Materials

Sodium alginate was provided by FMC International, Wallingstown, Ireland. Molecular weight is in the range of 90–180 kDa, and mannuronic content is in the range of 60% to 65%, with guluronic acid in the range of 35% to 40%. Hyaluronic acid was obtained from Acros Organics, Geel, Belgium, and has a molecular weight of around 2000 kDa. Calcium chloride and sodium chloride were obtained from Carl Roth, Karlsruhe, Germany. PCL (CAPA 6500) was a kind gift from Ingevity, Warrington, UK. Phosphate-buffered saline (PBS) was purchased from MP Biomedicals, Irvin, USA. Food coloring was a gift from Symrise, Holzminden, Germany. As a bioprinter, a Bioscaffolder 3.3 from Gesim was used, Radeberg, Germany. Gauge 25 conical nozzles and cartridges were obtained from Vieweg, Kranzberg, Germany. Calcein AM was obtained from AAT Bioquest, Pleasanton, USA, and ethidium homodimer I was obtained from ION Biosciences, San Marcos, USA. The cell culture medium keratinocyte growth medium (KGM) was purchased from Lonza, Basel, Switzerland. Dimethylformamide was purchased from VWR Chemicals, Radnor, USA. Methacrylic anhydride was purchased from Alfa Aesar, Haverhill, USA. Lithium phenyl (2,4,6-trimethylbenzoylphosphinat) (LAP) was obtained from BLDPharm, Shanghai, China.

2.1.1 | Hydrogel Development

HA and ALG were selected because of their advantages of being easily available and easy to handle. First, hyaluronic acid was dissolved in PBS without calcium ions and with food coloring for visualization. After being fully dissolved, ALG was added and gelled for at least 24 h. Until further use, the inks were stored at 4°C.

2.2 | Design of Experiment

For concentration and combination, a space filling design was chosen. HA (0–20 mg/mL) and ALG (0–32 mg/mL) were chosen as continuous factors. The concentration range was chosen beforehand with an inverse vial testing. The highest concentrations were chosen where the flow behavior was still visible.

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Both concentrations of zero were excluded from the experiment. For example, HA20ALG32 means HA with a concentration of 20 mg/mL and ALG with a concentration of 32 mg/mL.

For printing speed and pressure, a d-optimal design was chosen to exclude areas with high printing speed and low pressure and vice versa, as those areas could be determined to be not viable. Furthermore, neither the printing speed nor the pressure could be set to zero. All data analysis was done with Cornerstone Software (CamLine). In total, 68 experiments were conducted. An overview of each experiment can be found in the Supporting Information.

As a response in this model, printability in the form of image similarity, subjective ranking, and printability calculation was chosen. The DoE was evaluated separately for each response.

2.3 | Rheological Assessment

All rheological assessments were measured using a Haake RS6000 (Thermo Scientific) with a parallel plate geometry of 35 mm (PP-35) and a 650 µm gap. All measurements were carried out at 20°C, and before measurement, the hydrogel was let to rest and adjust for 240 s at 20°C. Data analysis was done by Haake RheoWin Software.

2.3.1 | Rotational Sweep

To understand the shear thinning behavior, a rotational sweep with an increasing shear rate ($\dot{\gamma}$) of 0 to 100 s^{-1} with 60 measure points was conducted. Four different regression models were examined with the RheoWin Data manager.

2.3.1.1 | Rheological Regression Models. To assess shear thinning behavior, four different regression models were used.

Ostwald de Waele or power law equation shown in Equation (1):

$$\tau = c * \dot{\gamma}^{p} \tag{1}$$

where τ is the shear stress, c (Pa s) is the consistency index, $\dot{\gamma}$ is the shear rate (s^{-1}) and p is the power law index.

The second regression model after Cross was calculated using Equation (2):

$$\frac{\eta(\dot{\gamma})}{\eta_0} = \frac{1}{1 + (c * \dot{\gamma})^{p}} \tag{2}$$

where c is the Cross constant (s) and p is the Cross exponent.

The third regression model after Carreau-Yasuda was calculated using Equation (3):

$$\frac{\eta(\dot{\gamma}) - \eta_\infty}{\eta_0 - \eta_\infty} = \frac{1}{\left(1 + \left(c * \dot{\gamma}\right)^{p_1}\right)^{\frac{1-p}{p_1}}} \tag{3}$$

where p_1 is the Yasuda exponent, c [s] is the relaxation time, and p is the power law index.

The last regression model after Carreau was calculated using equation (4):

$$\frac{\eta(\dot{\gamma})}{\eta_0} = \frac{1}{\left(1 + \left(c_1 * \dot{\gamma}\right)^{2}\right)^{p}} \tag{4}$$

where c_1 is the Carreau constant (s) and p is the Carreau exponent.

In all models is η the dynamic viscosity, η_0 the zero-shear viscosity and η_∞ the infinity shear viscosity.

2.3.2 | Oscillatory Amplitude Sweep

Oscillatory rheology was used for the amplitude sweep to determine the storage modulus G' and loss modulus G'' in the linear viscoelastic region (LVR). An amplitude sweep with a deformation rate of 0.1% to 500% was performed at a frequency of 1.592 Hz (= 10 rad/s) with 50 data points. The end of the LVR was calculated with the RheoWin Data software as the point where a 5% deviation from the plateau occurred.

2.3.3 | Oscillatory—Rotational—Oscillatory (ORO)

For the ORO in the first batch, oscillatory measurement was chosen with a deformation rate inside the LVR, with a frequency of 3.183 Hz over 60 s. As the second interval, a rotational shear rate of 100 s^{-1} was chosen. In the third section, the same adjustments as in the first section were chosen. After 15 and 180 s in the third section, the recovery in percentage was calculated with the RheoWin Data software.

2.4 | Printing Process

Before printing, the cartridge was loaded with each hydrogel and placed in the bioscaffolder. Two different printing methods were applied: pneumatic extrusion printing for hydrogel extrusion with a nozzle size of 250 µm and a drop-on-demand technique using piezoelectric ink jetting to drop the crosslinking solution (75 mg/mL $CaCl_2$ solution) onto the scaffold, leading to immediate crosslinking (Figure 1). The cartridge temperature was set at 20°C. Depending on the hydrogel, different printing speeds and pressures were chosen. The pressure range was between 15 and 100 kPa. The printing speed ranged from 1 to 75 mm/s.

As a printing design, a simple scaffold with a lattice was chosen purely to demonstrate printability and evaluate image similarity. The theoretical physical dimension of the scaffold was 7.5 × 7.5 × 0.4 (L/W/H) mm. An inner scaffold pattern with orthogonal alignment and a distance of 1.8 mm from strand to strand and a deposition of two layers was chosen (Figure 1).

2.5 | Printability Assessment

Printability assessment was done by image analysis. The images were taken with a stereomicroscope (Stemi 508) using an Axiocam 208 camera. On the one hand, the printed constructs

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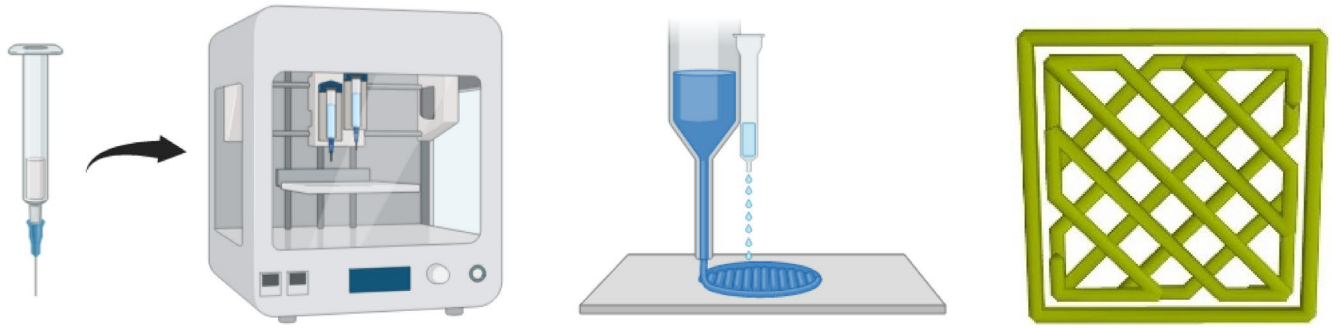

FIGURE 1 | Printing process with insertion of cartridge, temperature control, and extrusion and inkjet printing and CAD model of the simple scaffold.

were evaluated subjectively, and on the other hand, open-source deep learning image similarity was used to assess the printed hydrogel constructs. As a third method, the printability index using Equation (5) was calculated when possible.

$$Pr = \frac{L^2}{16 \times A} \tag{5}$$

with Pr as the printability index, L as the perimeter, and A as the area of a pore. Whenever applicable, the mean of 4 pores was calculated. As some printed scaffolds were unable to form pores and viable scaffolds, those were left as 0 in the DoE software.

For the subjective ranking, each printed scaffold was ranked according to printability.

An open-source Python script was modified and utilized here. To perform the comparison, a PCL scaffold with identical dimensions was printed and used as the baseline (set as 1). In addition, the original CAD image was included in the analysis. The difference between the original CAD model and the PCL-printed scaffold was added to each print value for normalization. For the image comparison, three different pretrained neural network architectures (ResNet50, VGG19, and EfficientNet) were selected. Each model was equally weighted in the analysis.

2.6 | Bioprint

2.6.1 | Synthesis of HAMA

To evaluate the model, a bioprint was conducted. For this, a different ink of another material was tested. Methacrylated hyaluronic acid (HAMA) was synthesized as previously described [24]. Briefly, high molecular HA (2000 kDa) was dissolved in bidistilled water and stirred at room temperature overnight. On the next day, dimethylformamide was added in a 1:2 DMF/water ratio. Methacrylic anhydride was added dropwise, while maintaining the pH between 8 and 9 with NaOH. After stirring under light exclusion for 20 h, the pH was adjusted to 7 and further diluted with water. Sodium chloride was added to obtain a final concentration of 0.6 M. Finally, the solution was transferred to the dialysis membrane (Spectra Por 6–8 kDa cut-off) and dialyzed against demineralized H_2O for 10 days at 4°C, followed by lyophilization. HAMA was stored at −20°C until further use.

2.6.2 | Cell Culture

As cells, an SV40-immortalized cell line of human corneal keratocytes (HCK) was used. The cell line was a kind gift of Dr. M. Zorn-Kruppa and was also characterized by Zorn-Kruppa et al. [25] The HCK cells were cultured in KGM supplemented with $CaCl_2$ at a final concentration of 0.5 M in 37°C and 5% CO_2 at a humified atmosphere.

2.6.3 | Rheology

The same rheological tests were conducted as previously described. Due to the circumstance of less material availability, the plate geometry was swapped with a plate cone geometry with a 1° angle and a diameter of 35 mm.

2.6.4 | Recreation of Bioink

The final concentration of HAMA was 30 mg/mL, further known as HAMA30. For crosslinking, LAP as the photoinitiator was added to achieve a final concentration of 1.32 mmol/L. HCK cells were added before printing to obtain the final concentration of 10^6/mL. The printing process was conducted as previously described. As HAMA relies on crosslinking with UV light, after printing, each layer was crosslinked with a 405 nm UV lamp mounted on the print head.

2.6.5 | Live/Dead Staining

The samples were stained using ethidium homodimer I (2 μM) and calcein-AM (5 μM). The stained samples were observed using a fluorescence microscope (Olympus IX50) and images (Olympus DP28) were taken. For the calculation of cell viability, images of three randomly chosen areas of the samples were taken, and viability was calculated as follows:

$$\text{Cell Viability}\,(\%) = \frac{\text{Live Cells}}{(\text{Live Cells} + \text{Dead Cells})} \tag{6}$$

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Live/Dead staining was performed on day 0, 1, 2, 4 and day 7.

2.7 | Statistics

Data were analyzed with one-way analysis of variance (ANOVA) with a Tukey post hoc test using OriginLab 2024b software. Prior to ANOVA, data were checked for outliers using Grubbs' test and for normal distribution using the Shapiro–Wilk test, and homoscedasticity was assessed with Levene's test. A significance level of $*p < 0.05$ was used. Data are presented as mean ± standard deviation. Unless otherwise stated, all measurements were at least conducted as triplicates.

3 | Results and Discussion

3.1 | Rotational Sweep

All hydrogels exhibited shear thinning behavior shown in Figure 2A. An increase in concentrations led to higher viscosity in general, while this effect was more pronounced with HA than with ALG in the mixture. It has been stated that shear thinning is key for extrusion bioprinting [26]. Nevertheless, not all hydrogels demonstrated high printability; thus, shear thinning is not sufficient to prove printability. Therefore, a look at the zero-shear viscosity can be crucial to better understand the behavior between printability and rheological characteristics. The zero-shear viscosity is the calculated viscosity of the ink at the lowest shear rate, which is zero. Thus, this can be seen in the property of the ink at rest before extrusion printing started.

3.2 | Rheological Regression Models

Rheological regression models can be useful to describe the flow behavior of fluids and gels. As the power law model is a rather simple model describing shear thinning or shear thickening behavior, models such as Cross, Carreau–Yasuda, and Carreau consider more complex approaches with the inclusion of zero-shear viscosity.

All power law indices (p) of the Ostwald de Waele regression model were between 0.23 and 0.79. A power law index below 1 is an indicator of shear thinning behavior. Thus, we further proved that all hydrogels demonstrated shear thinning behavior. The consistency index (c) was between 1.12 ± 0.19 Pa s and 357.94 ± 74.81 Pa s, growing with HA and increasing concentration. This mathematical rheological regression model showed a high correlation between 0.92 and 1.00. Unlike the other regression models, with the power law model, the zero-shear viscosity is not calculated. Instead of the zero-shear viscosity, the consistency index is calculated. The consistency index is more a measurement of the thickness and resistance of a fluid to flow. Therefore, other regression models such as Cross, Carreau, and Carreau-Yasuda are more suitable.

The Cross and Carreau-Yasuda regression models showed a similar high correlation between 0.95 and 1.00. Although they expressed similar trends with regard to zero-shear viscosity, with increasing HA and higher concentration, the zero-shear viscosity grew; the values of zero-shear viscosity differ with the addition of HA to the system, as can be seen in Figure 2C,D. Both values for ALG were similar, but not for HA. For example, the lowest zero-shear viscosity for ALG16 reached 0.74 ± 0.07 Pa s and 0.66 ± 0.07 Pa s in the Cross and Carreau-Yasuda regression model, respectively. For the gel with the highest content, HA20ALG32, zero-shear viscosities reached 312.80 ± 86.48 Pa s and 151.11 ± 24.40 Pa s in the Cross and Carreau-Yasuda regression model, respectively. Thus, the zero-shear viscosity had a steeper increase with the Cross regression model than with the Carreau-Yasuda regression model. For reasons of clarity, the significant statistics are not shown here but can be found in the Supporting Information.

The correlation coefficient of the Carreau model was, except for the ALG gels, very poor and thus is not discussed further, as this model was unfit for determining the flow behavior of these gels. An overview of each regression model can be found in Table 1.

3.3 | Oscillatory Amplitude Sweep

The storage modulus G' describes the elastic behavior of a material, thus the solid-like properties of the hydrogel and shape retention. This is important for the bioprinting process, as an optimum can be determined for extrusion printing. A storage modulus that is too high leads to overgelling and thus to poor printability. A G' that is too low means that the material flows too easily and therefore cannot be extruded. Nevertheless, G' should not be considered alone, the loss modulus G'', which describes the viscous behavior of a material and thus the liquid-like properties of the hydrogel, is also important. The ratio of G'' to G', commonly known as damping factor or loss tangent/factor ($\tan\delta$), indicates sol-like behavior ($\tan\delta > 1$), or gel-like behavior ($\tan\delta < 1$). It has been stated that a damping factor between 0.25 and 0.6 is correlated with improved printability [19].

A similar trend to the zero-shear viscosity can be observed in the measurement of the storage modulus. An increase in content and the addition of HA lead to the rise of G' (Figure 2E). For reasons of clarity, the significant statistics are not shown here but can be found in the Supporting Information. Although for the damping factor no considerable differences could be seen between the HA hydrogels, only a significant difference could be seen from ALG16 and ALG32 to the other hydrogels. The damping factor was in the range of 0.45 and 0.98 for HA20 and HA20ALG32, respectively (Figure 2F). These data now suggest that the damping factor has little influence on printability, and as will be explained later, not all gels depicted similar printability though having similar damping factors. Nevertheless, the claim that a damping factor below 1 is crucial for printing can be made.

3.4 | Oscillatory–Rotational–-Oscillatory (ORO)

An ORO recovery test is a common method to simulate the extrusion process, during which the gel transitions from its initial state to high pressure and then again rests. To display the high pressure by printing, a shear rate of 100 s^{-1} was chosen, as this

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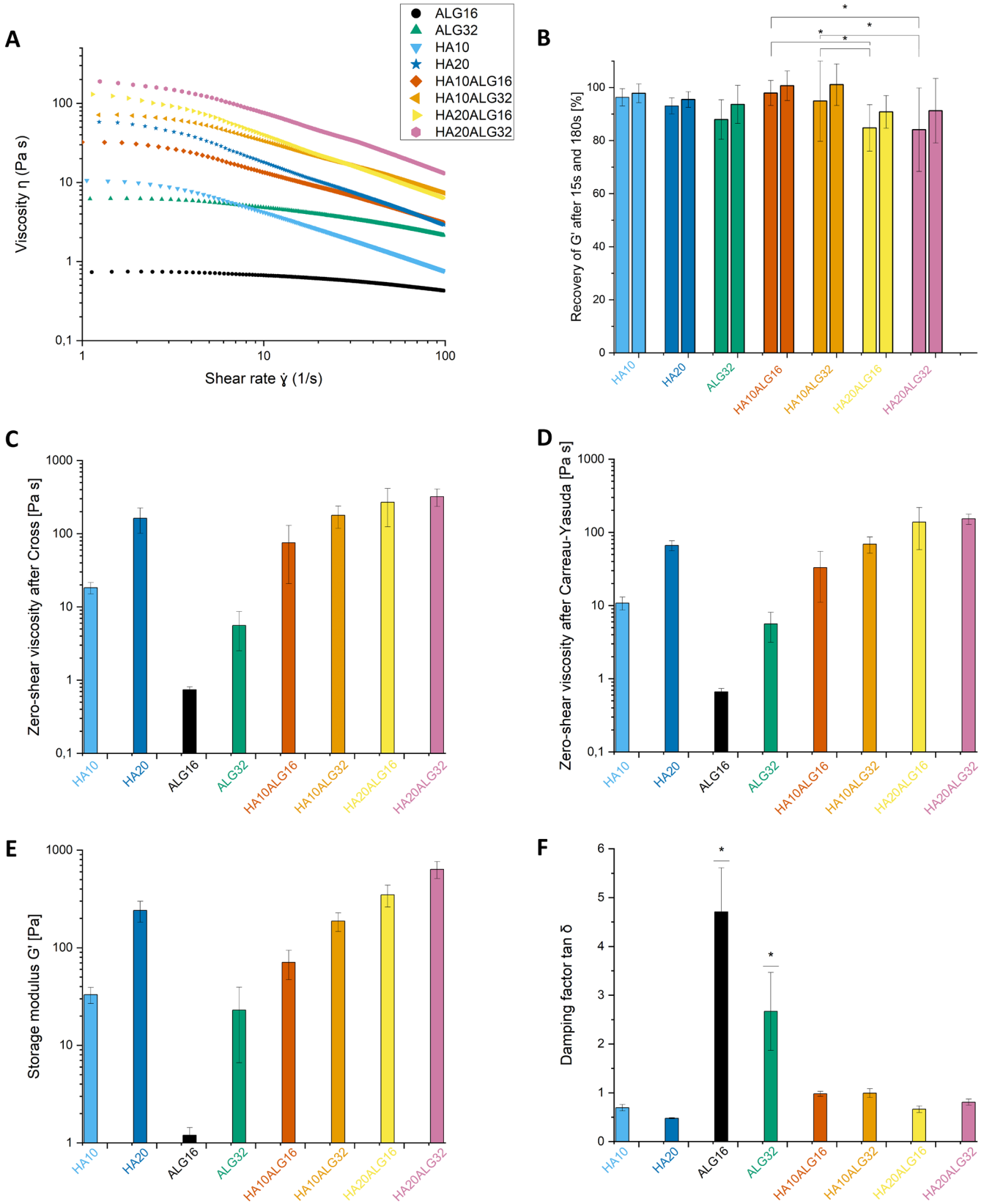

FIGURE 2 | (A) rotational shear thinning behavior of hydrogels; ALG16 ($n = 9$), ALG32 ($n = 6$), HA10 ($n = 9$), HA20 ($n = 8$), HA10ALG16 ($n = 9$), HA10ALG32 ($n = 9$), HA20ALG16 ($n = 16$), HA20ALG32 ($n = 9$), Note, for clarity, graphs are displayed without standard deviations. (B) Recovery of G' after 15 s (first column) and 180 s (second column); HA10 ($n = 9$) HA20 ($n = 9$), ALG32 ($n = 9$), HA10ALG16 ($n = 9$), HA10ALG32 ($n = 6$), HA20ALG16 ($n = 12$), HA20ALG32 ($n = 9$). (C, D) Zero-shear Viscosities after Cross and Carreau-Yasuda, respectively; ALG16 ($n = 9$), ALG32 ($n = 6$), HA10 ($n = 9$), HA20 ($n = 8$), HA10ALG16 ($n = 9$), HA10ALG32 ($n = 8$), HA20ALG16 ($n = 14$), HA20ALG32 ($n = 9$). (E, F) Storage modulus G' and damping factor, respectively; ALG16 ($n = 8$), ALG32 ($n = 7$), HA10 ($n = 9$), HA20 ($n = 11$), HA10ALG16 ($n = 8$), HA10ALG32 ($n = 9$), HA20ALG16 ($n = 15$), HA20ALG32 ($n = 9$). Data is presented as mean ± standard deviation, graphs of each individual point can be accessed in the Supporting Information.

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TABLE 1 | Summary of rheological regression models for each bioink.

	ALG16	ALG32	HA10	HA20	HA10ALG16	HA10ALG32	HA20ALG16	HA20ALG32
Ostwald de Waele								
Consistency index c	1.12 ± 0.19	10.84 ± 4.17	22.17 ± 4.45	104.52 ± 12.57	58.83 ± 28.62	142.09 ± 46.20	208.59 ± 89.88	357.94 ± 74.81
Power law index	0.79 ± 0.04	0.66 ± 0.04	0.28 ± 0.03	0.23 ± 0.01	0.37 ± 0.03	0.37 ± 0.02	0.26 ± 0.08	0.30 ± 0.03
r	1.00	1.00	0.99	0.98	0.99	0.99	0.92	0.97
Cross								
Cross constant (s^{-1})	110.34 ± 57.72	122.14 ± 251.43	2.52 ± 0.44	1.01 ± 0.56	1.61 ± 1.01	1.33 ± 0.53	1.41 ± 0.83	2.11 ± 1.07
Cross exponent	0.81 ± 0.20	0.73 ± 0.31	0.86 ± 0.06	0.83 ± 0.05	0.70 ± 0.06	0.66 ± 0.03	0.82 ± 0.18	0.76 ± 0.06
Zero Shear Viscosity (Pas)	0.74 ± 0.07	5.58 ± 3.06	18.29 ± 3.23	162.87 ± 61.65	75.33 ± 54.55	178.74 ± 60.65	270.09 ± 145.57	321.28 ± 86.48
r	1.00	1.00	1.00	1.00	1.00	1.00	0.95	0.98
Carreau								
Carreau constant	221.95 ± 147.60	211.70 ± 101.00	164.85 ± 65.29	58.67 ± 3.93	205.93 ± 17.61	205.83 ± 5.87	111.56 ± 91.31	142.44 ± 17.62
Carreau exponent	2.08 ± 1.57	2.86 ± 1.39	4.12 ± 1.57	1.34 ± 0.28	4.63 ± 0.44	4.56 ± 0.07	2.99 ± 1.65	3.40 ± 0.44
Zero Shear Viscosity (Pas)	0.57 ± 0.04	3.21 ± 1.89	2.18 ± 0.59	11.86 ± 1.66	7.00 ± 2.74	16.91 ± 4.59	19.80 ± 7.82	41.54 ± 7.49
r	0.98	1.00	0.56	−0.1	0.87	0.88	0.48	0.64
Carreau-Yasuda								
Yasuda exponent	2 ± 0.00	2.00 ± 0.00	2.00 ± 0.00	2.00 ± 0.00	2.00 ± 0.00	2.00 ± 0.00	2.00 ± 0.00	2.00 ± 0.00
Relaxation time	0.02 ± 0.00	0.00	0.04 ± 0.00	0.05 ± 0.25	−0.05 ± 0.41	−0.09 ± 0.26	0.09 ± 0.34	0.05 ± 0.00
Power law index	0.74 ± 0.04	0.59 ± 0.05	0.24 ± 0.03	0.20 ± 0.01	0.35 ± 0.03	0.34 ± 0.03	0.23 ± 0.10	0.26 ± 0.03
Zero Shear Viscosity (Pas)	0.66 ± 0.07	5.62 ± 2.47	10.85 ± 2.18	66.39 ± 10.34	33.06 ± 22.01	68.13 ± 17.27	133.16 ± 78.02	151.11 ± 24.49
r	1.00	1.00	1.00	1.00	1.00	1.00	0.95	0.98

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shear rate exhibited the lowest measured viscosity, as can be seen in Figure 2B. 15 s were chosen, as this is approximately the time needed to print one layer. To better understand the ability to rebuild its structure after high shear, recovery after 180 s was measured. The structural recovery of a non-Newtonian fluid over time is called thixotropy. Thixotropy is important, as it is desired that the gel recovers its original state after extrusion printing [26].

All hydrogels presented high to very high recovery, achieving over 80% recovery within 15 s, which further increased by 180 s. For each hydrogel, the recovery percentage improved over time, indicating that the hydrogels exhibit thixotropic behavior. This suggests that the hydrogels are capable of rapid recovery and the ability to retain their original structure after extrusion. However, none of the hydrogels fully regained their initial G' values even after 180 s, which hints that either an irreversible change, rheodestruction, has occurred or that a longer time is required for the hydrogels for complete structural recovery. With increasing content of HA and ALG, a slight decline in recovery of G' could be observed, as the mean difference between HA10ALG16 15 s and HA20ALG16 15 s and HA20ALG32 15 s is significant. Furthermore, the mean difference between HA10ALG32 15 s and HA20ALG16 15 s and HA20ALG32 15 s is also significant. ALG16 was not suitable for the ORO, thus no data could be obtained and it is not included here.

3.5 | Design of Experiment

In total, 68 runs with seven different hydrogel concentration/combinations were done. Depending on the hydrogel, different runs of printing speed and pressure were conducted. For example, for ALG16, three runs were performed, while for HA20ALG32, 15 runs were carried out. Out of the 68 runs, 42 were viable for printability index calculation. There are limits to the printability index, as it can only be used if pores are present. In the case of more complex models or very poor printability, the calculation cannot be used. Here, 26 runs had such low printability that they had to be excluded from the DoE input data and left as 0. For subjective ranking, each picture was first classified into different groups, followed by a sub ranking. For the image similarity scoring, each picture was not further modified. An example of high printability, over extruded and under extruded can be seen in (Figure 3). An overview of each picture with the according ranking can be found in the Supporting Information.

Deep learning image similarity looks at edges, textures, shapes, and patterns to compare images [27]. In this case, shape and pattern are particularly important. The same dye was chosen to reduce the influence of the color and emphasize the actual shape of the printed object. Nevertheless, because of the opaque nature of deep learning models, it is unclear how they weight different features when determining similarity. Thus, a comparison of the image similarity models with a subjective evaluation by a human is required to check the correspondence with human perception. Therefore, we decided to evaluate the deep learning image similarity by comparing the outcome of the predicted DoE models (Figure 4). Especially, the outcome of the image similarity scoring and the subjective ranking was remarkably similar. Although the predicted graph of the calculated printability is slightly different, the tendencies stay the same as for all graphs: With increasing ALG concentration, the printability decreases; with growing HA concentration, the printability improves; and with greater pressure and printing speed, the printability declines. Therefore, deep learning image similarity can be an alternative to the printability index as it considers more viable data and is not as dependent on the pore size. Additionally, it is on a similar level to a subjective ranking while it takes less time. We propose that deep learning image similarity can be a valuable tool to evaluate printability. Furthermore, in this study, pretrained data was used. A deep learning model trained with bioprinted data could have the potential to be

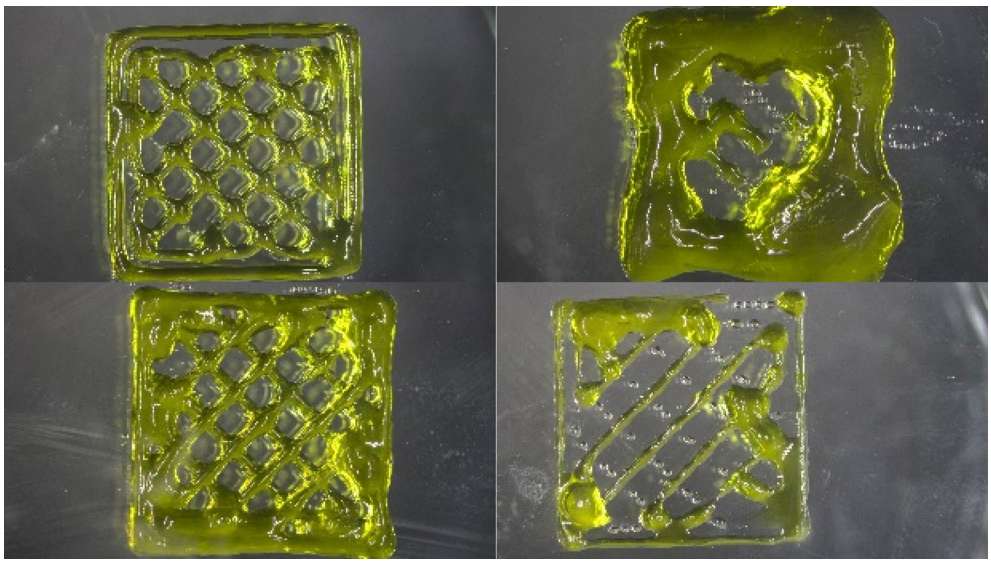

FIGURE 3 | Printed scaffolds with very good printability (top left), over extruded (top right) and good printability (bottom left) and under extruded (bottom right).

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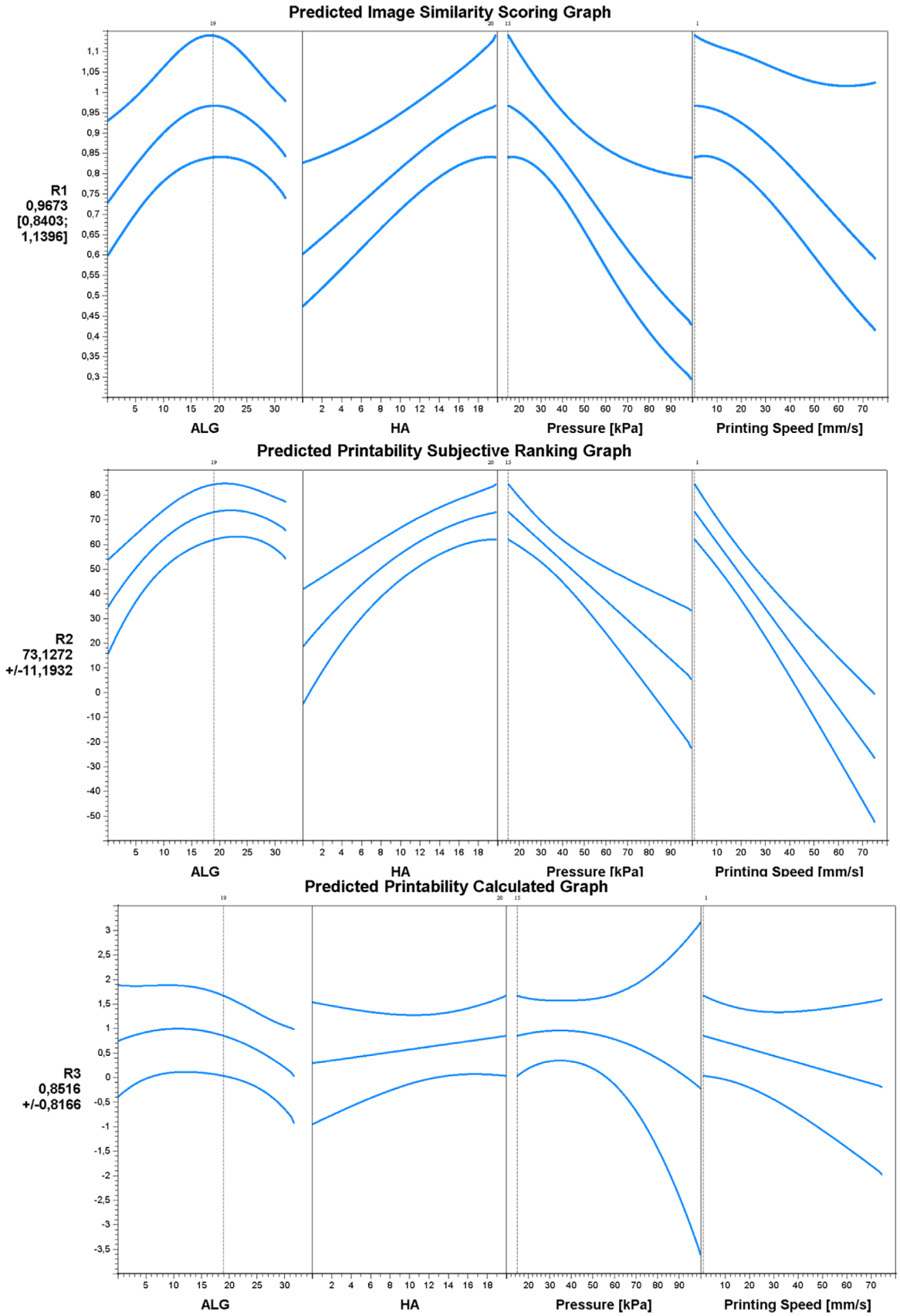

FIGURE 4 | Predicted graphs for optimal setting to gain highest printability according to printability image scoring, subjective image ranking, and printability index.

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TABLE 2 | Printability-coefficient table (after image similarity scoring) with coefficients, standard error, *T*-value, and significance.

Term	Coefficient	Standard error	*T*	Significance
Constant	0.8700	0.0118	73.9997	< 0.0001
ALG	0.0282	0.0094	2.9878	0.0043
HA	0.0608	0.0137	4.4348	< 0.0001
Printing Speed	0.0063	0.0074	0.8548	0.3967
Pressure	−0.0612	0.0141	−4.3256	0.0001
ALG × HA	−0.1194	0.0179	−6.6620	< 0.0001
ALG × Printing Speed	−0.0852	0.0164	−5.1856	< 0.0001
ALG × Pressure	0.1271	0.0206	6.1669	< 0.0001
HA × Printing Speed	−0.0832	0.0169	−4.9196	< 0.0001
HA × Pressure	0.1231	0.0269	4.5744	< 0.0001
Pressure × Printing Speed	0.0939	0.0180	5.2312	< 0.0001
ALG^2	−0.0761	0.0119	−6.3810	< 0.0001
HA^2	−0.0472	0.0117	−4.0457	0.0002
Printing $Speed^2$	−0.0363	0.0097	−3.7306	0.0005
$Pressure^2$	−0.0566	0.0136	−4.1582	0.0001

even more accurate. Nevertheless, all information from now on will be discussed solely on the image similarity scoring results.

The calculated model can be described using the associated terms, which are shown in Table 2.

Considering decent printability made by the color range of blue, only HA10ALG32, HA20ALG16, and HA20ALG32 demonstrated printability (Figure 5). The highest printability was achieved by HA20ALG16. The model predicts the highest printability with a HA20ALG19 hydrogel. Poor printability could be seen in all single hydrogels, HA10, HA20, ALG16, and ALG32, as well as in HA10ALG16. Thus, the combination as well as the higher content of a component leads to higher printability. Looking back at the rheology data, the damping factor and the recovery rate could not be used to make statements about the corresponding printability. The difference between the hydrogels could be seen in the zero-shear viscosity and in G'. Therefore, a zero-shear viscosity between 16.91 ± 4.59 Pa s and 41.54 ± 7.49 Pa s or between 68.13 ± 17.27 Pa s and 151.11 ± 24.49 Pa s, according to the Carreau or Carreau-Yasuda regression model, respectively, is desirable. A G' between 182.14 ± 16.94 Pa and 617.67 ± 56.43 Pa is in the range of being printable. Although HA20 with a G' of 253.51 ± 16.64 Pa is also in this range, it showed poor printability and was not suitable. Therefore, considering only one rheological characteristic such as G' is not sufficient to assess printability. This DoE model also predicted that there is a linear trend between printing speed and pressure. Lower printing speed and lower pressure lead to improved printability. As often stated, high pressure is correlated with high shear stress, leading to more cell death in the bioprinting field [28–30]. This data here suggests the use of low printing speed with low pressure to maximize printability and therefore allow less shear stress and thus possible greater cell viability.

3.6 | Bioprint

To evaluate the model and utilize the information for actual bioprinting, a bioprint was performed with a single component gel consisting of HAMA at a higher concentration than the HA concentration previously used to ensure that the rheological data was within the range of the gels that were considered printable. Similar to the mentioned hydrogels, HAMA30 also proved shear thinning behavior. Regarding zero-shear viscosity, HAMA30 showed a higher value with 224.86 ± 87.21 Pa s of the Cross model than HA10ALG32. With a zero-shear viscosity of 124.36 ± 43.49 Pa s of Carreau-Yasuda models, it had a slightly lower zero-shear viscosity than HA20ALG16. By comparing the G' value, 405.92 ± 92.65 Pa, HAMA30 ranked between HA20ALG16 and HA20ALG32. Thus, printing speed and pressure were adjusted to an area between the predicted HA20ALG16 optimum and the predicted HA20ALG32 optimum. Two different settings were set: A printing speed of 1 mm/s and 25 kPa, as well as a printing speed of 2.5 mm/s and 30 kPa. Both settings demonstrated high printability according to this model (Figure 6). Therefore, we conclude that prediction models and contour plots can be used to adjust printing speed and pressure if rheological properties such as zero-shear viscosity and G' are known.

Both bioprinted constructs exhibited high viability over the course of 7 days (Figure 6). Optimal cell viability is vital for many applications, for example, bioprinting of tissues or for the use of in vitro models [31]. No significant difference can be seen between the constructs (Figure 6). Thus, since printability is very similar, it is recommended to use the higher printing speed for time-saving reasons. HAMA is a commonly used biomaterial in the bioprinting field [10]. HAMA, in general, displays a superior transparency, making it suitable for corneal bioprinting [32]. In terms of

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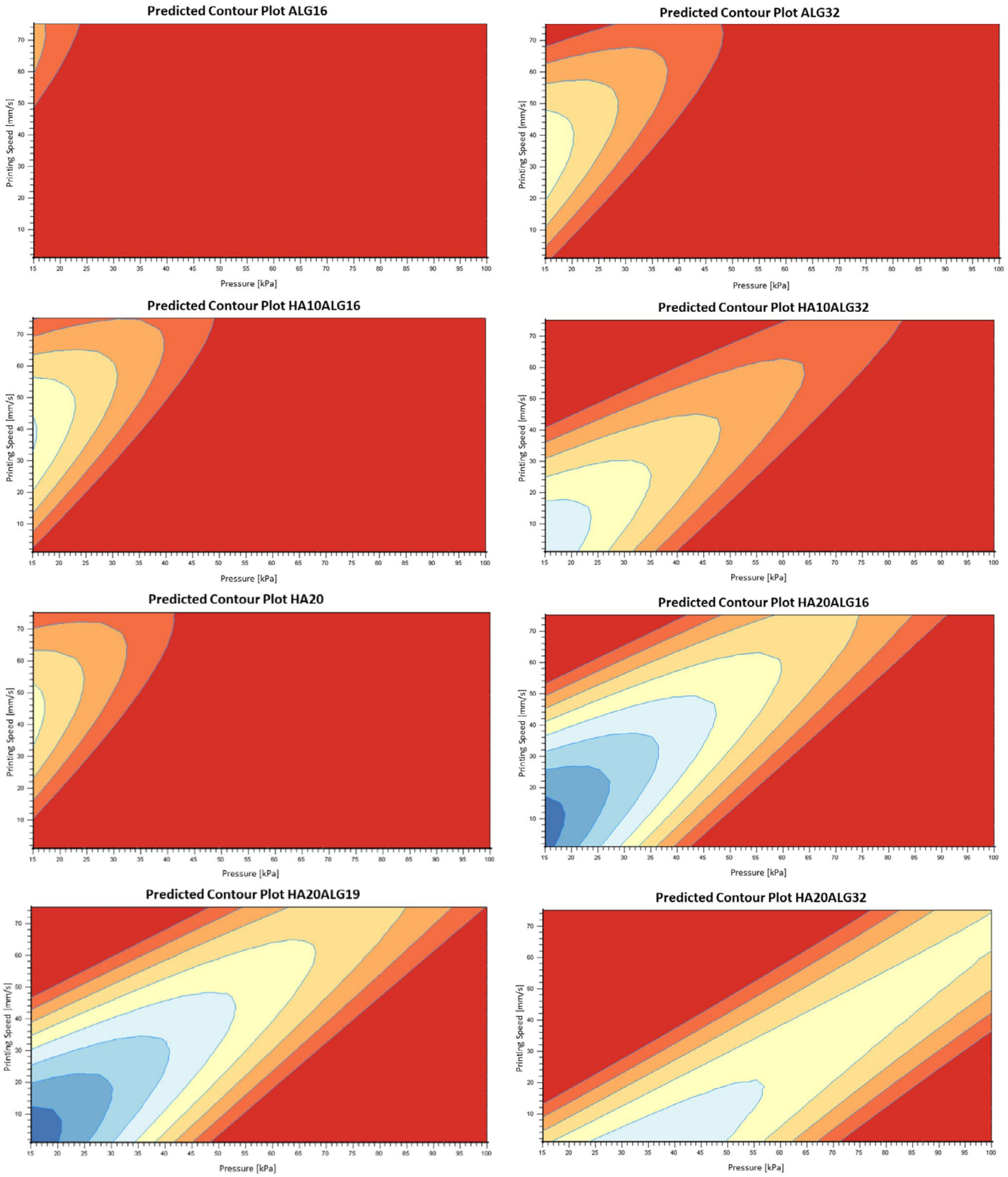

FIGURE 5 | Contour plots of each printed hydrogel as well as optimized hydrogel (HA20ALG19), from blue to red printability decreases.

bioprinting corneal models, HA and modified HA have been utilized, but HAMA has not yet been implemented to our knowledge [3, 33]. Rheological characterization of bioprinted hydrogels for corneal bioprinting was not done in this extent. Shear thinning behavior was often investigated or viscoelastic properties such as G' were examined [34–36]. This rheological data on HAMA30 and printability provides further insight into possible biomaterials suitable for corneal bioprinting. We acknowledge that this study presents a small step in the direction of corneal bioprinting. The broader goal of corneal bioprinting needs further studies on different characterization methods such as transparency, mechanical properties, and biocompatibility. Although this study involves HCKs, the overall strategy developed here can be applied to a broader bioprinting application. The information gained from the

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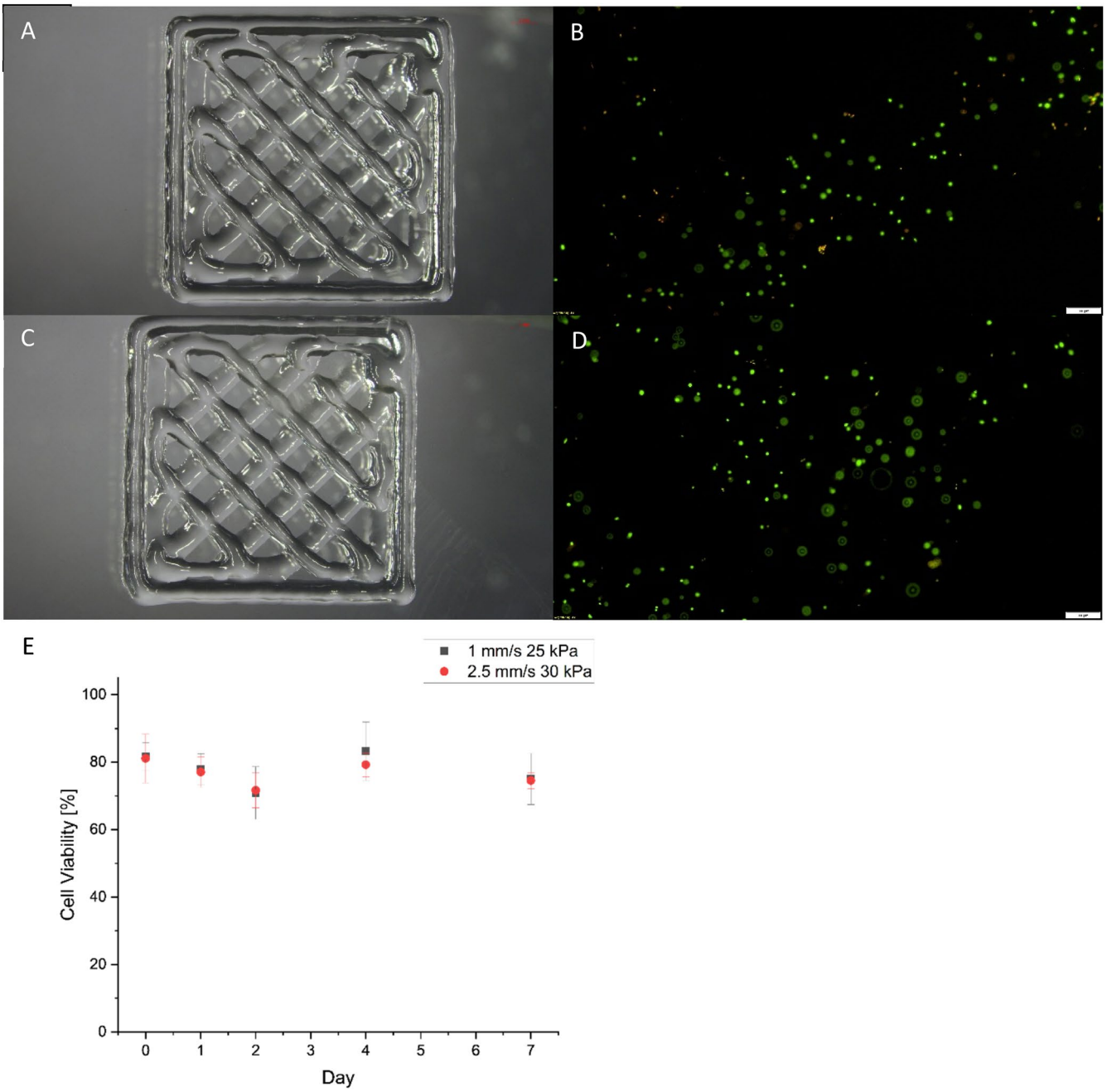

FIGURE 6 | (A) Printed model of 1 mm/s and 25 kPa. (B) Live/Dead staining of printed model of 1 mm/s and 25 kPa. (C) Printed model of 2.5 mm/s and 30 kPa. (D) Live/Dead staining of printed model of 2.5 mm/s and 30 kPa. (E) Cell viability over the course of 7 days.

DoE offers a transferable framework suitable for evaluating and optimizing a wide range of bioinks for extrusion-based bioprinting.

4 | Conclusion

This study demonstrated a new method to evaluate printability with the help of deep learning image similarity. We propose that in the future, the use of deep learning, in general, can be a beneficial tool to the bioprinting community. Currently, the use of the deep learning image similarity here was only of the top view of the printed constructs, although it is not limited. We suggest that further use of images from different angles could even improve this model and can be used on more complex models instead of the current lined pattern scaffolds.

Furthermore, we were able to understand the influence of rheological characteristics as well as printing speed and pressure on printability. This design of experiment hints that optimal printing can be achieved with low printing speed and pressure. In regards to rheological characteristics, this data suggests that looking at only one property is not sufficient to understand printability. In particular, the combination of oscillatory and rotational methods can enhance the understanding of hydrogels. A minimum of zero-shear viscosity and a storage modulus seem to be needed to be printable. Unlike what other research suggests, only a damping

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factor in a specific range does not guarantee high printability. Although quick recovery is welcomed to regain the original state, a lower recovery does not correlate to a decrease in printability.

In addition, the evaluated hydrogels can be used as surrogate hydrogels with certain rheological properties. We have proven that the rheological data and the predicted models by the DoE can be transferred to another hydrogel with a different crosslinking method. We used high molecular HAMA with the combination of HCKs to establish the first, foundational step for further corneal bioprinting. Future work will focus on more investigations into corneal bioprinting based on the rheological properties of the bioink presented here.

This work is a further advance in the understanding of extrusion bioprinting, rheology, and printing parameters. It can be used as a guide for the production of novel bioinks with optimum printability from a rheological point of view.

Author Contributions

Leon Balters and Stephan Reichl did the conceptualization and design of the study. Leon Balters did the visualization, methodology, investigation, data curation, and writing of the original draft. Stephan Reichl did the supervision, resources, project administration, and writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All data or related information supporting the conclusions of this study is included in the article, Supporting Information, or can be obtained separately.

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Supporting Information

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